

Vulnerability Assessment Report

23rd July 2026

System Description

The server hardware consists of a powerful CPU processor and 128GB of memory. It runs on the latest version of Linux operating system and hosts a MySQL database management system. It is configured with a stable network connection using IPv4 addresses and interacts with other servers on the network. Security measures include SSL/TLS encrypted connections.

Scope

The scope of this vulnerability assessment relates to the current access controls of the system. The assessment will cover a period of three months, from June 2026 to August 2026. [NIST SP 800-30 Rev. 1](#) is used to guide the risk analysis of the information system.

Purpose

The remote database server provides value by serving as an organized collection of information that manages large amounts of critical data and services on a network such as analytic, customer, and other data. Securing this data is essential for maintaining the confidentiality and integrity of sensitive personal information which helps the business comply with legal regulations. If the server were disabled, the business would suffer from interrupted production and services, potentially leading to large financial damages and lost productivity. Furthermore, a prolonged service outage could permanently damage the organization's reputation, causing customers to lose trust and seek out competitors instead.

Risk Assessment

Threat source	Threat event	Likelihood	Severity	Risk
Hacker	Altering or deleting critical database records to disrupt operations along with viewing personal information.	3	3	9
Competitor	Perform espionage to collect	2	3	6

	<i>sensitive data for competitive advantage</i>			
<i>Customer</i>	<i>Unauthorized modification of order/account data for financial gain</i>	<i>2</i>	<i>3</i>	<i>6</i>

Approach

A risk assessment based on NIST SP 800-30 Rev. 1 was utilized to evaluate threats associated with the publicly accessible database. The three threat sources: Hackers, Competitors, and Customers, were selected because a database that's open to the public presents an unauthenticated attack surface to any external entities. Hackers pose a critical risk of harming operations or stealing data, while competitors focus on data espionage for competitive advantage. Additionally, malicious customers pose the risk of tampering with data for financial gain. The probabilities were then weighed against potential operational impact to ensure an accurate evaluation of overall risk.

Remediation Strategy

To remediate and mitigate these identified risks, the organization should implement an Authentication, Authorization, and Auditing framework to enforce the principle of least privilege. Enforcing Role-Based Access Control alongside Multi-Factor Authentication will prevent unauthorized users from gaining administrative privileges to the database. To secure the network, a firewall should be configured to block all random connections from the public internet and only allow the specific IP addresses of authorized employees, effectively creating a controlled zone. Finally, securing data in transit using TLS encryption will protect sensitive analytics and customer records from exfiltration.